



AI-Based Crop Management & Disease Advisory System

Presented by: Aryan Chourasia

Roll No.: 2501320100070

Greater Noida Institute of Technology

Guided by: Khushi Rattan

Problem Statement

Challenges in Crop Disease Management

- Crop diseases can reduce yield and crop quality.
- Early identification is difficult without expert support.
- Farmers may struggle to understand treatment and prevention options.
- A simple AI-assisted advisory system can provide faster, understandable guidance.

Objectives

What Verdant Aims to Achieve

- Provide an easy crop-image upload and analysis interface.
- Identify crop disease through an image-based prediction module.
- Show prediction confidence and severity.
- Generate practical AI-based treatment and prevention advice.
- Make agricultural guidance easier to understand and act upon.

Proposed Solution

How Verdant Works

- User uploads a clear crop or leaf image.
- The disease prediction module analyzes the image.
- The system obtains crop, disease, confidence and severity information.
- Groq-powered AI generates personalized crop-management advice.
- Results are displayed through an interactive web interface.

Key Features

Core Capabilities

-  Crop/leaf image upload and preview
-  Disease prediction interface
-  Confidence and severity display
-  AI-powered agricultural advisory
-  Organic treatment suggestions
-  Prevention and crop-risk guidance
-  Clear and analyze another image

AI Advisory

Six Questions Answered by Verdant

- 1. What caused this?
- 2. How can I treat it?
- 3. Can I use an organic treatment?
- 4. How often should I apply the treatment?
- 5. How can I prevent it next season?
- 6. Is this dangerous for my whole crop?

Technology Stack

Technologies Used

- Frontend: HTML5, CSS3 and JavaScript
- Backend: Node.js HTTP server
- AI: Groq API and Groq SDK
- Data exchange: JSON
- Development: Visual Studio Code
- Environment: Local Node.js application

System Workflow

End-to-End Architecture

- Crop Image → Upload & Preview → Disease Prediction
- Disease + Crop + Severity → Backend API
- Backend → Groq AI → Structured JSON Advisory
- Frontend → Interactive six-question advisory
- User → Treatment, prevention and risk information

Current Prototype

Demonstration Status

- The current interface demonstrates the complete user experience.
- The prototype can display a sample prediction such as:
 - Tomato — Septoria leaf spot
 - 96% confidence · Severe
- The next integration step is connecting the upload interface to the actual trained disease-classification model.

Benefits

Why Verdant Is Useful

- Faster access to understandable crop-management information.
- Combines prediction results with actionable AI guidance.
- Simple interface suitable for non-technical users.
- Supports preventive thinking, not only disease identification.
- Can be extended to more crops, diseases and languages.

Future Scope

Planned Improvements

- Connect a real image-classification model.
- Support more crops and disease classes.
- Add prediction history and crop-health dashboards.
- Integrate weather-aware recommendations.
- Add multilingual and mobile/PWA support.
- Explore location-based agricultural guidance.

Safety & Limitations

Responsible Use

- AI advice should be treated as guidance, not a replacement for agricultural experts.
- Treatment decisions should consider product labels and local agricultural guidance.
- The system should avoid inventing chemical doses.
- Prediction accuracy depends on image quality and the underlying model.
- Professional verification is recommended for uncertain or serious cases.

Conclusion

Verdant: Detect • Understand • Treat • Prevent

- Verdant combines image-based crop disease detection with an AI agricultural advisor.
- It turns a prediction into practical, easy-to-understand next steps.
- The project provides a foundation for a scalable smart crop-management platform.

THANK YOU

Questions & Discussion

Aryan Chourasia

Roll No.: 2501320100070

Greater Noida Institute of Technology

Guided by: Khushi Rattan